

CĂLIN ILIE

calin@ilie.io
07776900428

github.com/calini
linkedin.com/in/calini

SKILLS

Computer Languages	Go, Python, JS/TS, Java/Kotlin
Infrastructure	Docker, Kubernetes, Helm, GKE, AWS, CI/CD
Other	Neo4j, SQL, RabbitMQ, Grafana, Spring Boot, React, HTMX

EXPERIENCE

Main Platform Refute - *Software Engineer* Jan 2025 - Present

- Led the development of the Main Platform from the ground up, working to product requirements
- Go + Templ + HTMX stack, chosen over a heavier SPA architecture to favour simplicity and server-side rendering
- Aggregates and visualises Neo4j graph data through filtering across themes, topics, narratives, and actor/group behaviour patterns, with closed-loop analyst labelling system feeding back into ML models and heuristics to continuously refine behavioural predictions
- Dual-purpose: analyst-facing frontend and a REST API designed for external client integration
- Auth0 authentication and a role-based permissions model
- Also maintained real-time data ingestion pipelines integrating third-party scraping sources (WebZ.io, Data365.co) via Redpanda Connect into RabbitMQ with error handling and DLQ management

CBE - Core Systems *bet365* - *Software Engineer* Jun 2022 - Sep 2023

- Maintained and improved a distributed bet recalculation engine: a high-throughput stream processing system underpinning instant cashout functionality across 10 million live bets, built in Erlang.
- Investigated and resolved redundant odds subscription logic, reducing memory footprint by 15% while simplifying system complexity and reducing active subscription count
- Implemented US tax calculation, extending the existing framework to handle per-state bracket rules and edge cases - designed to accommodate ongoing legislative changes
- Delivered a broad range of reliability, performance, and feature tickets across a 100k+ LoC codebase, including end-to-end changes through the full system

COS - Cashout Services *bet365* - *Software Engineer* Oct 2023 - Nov 2024

- Joined an in-progress Go microservices rewrite of the CBE engine, contributing across roughly half of the 10-service cluster.
- Designed and implemented a Docker-based local development environment for a previously near-impossible-to-run-locally 10-service cluster - including Compose orchestration and CI/CD integration, enabling developers to run any subset of services in debug mode while connecting seamlessly to the remainder in a shared dev environment
- Wrote mock injection tooling to simulate requests from services outside the local cluster, adopted across the team
- Improved test coverage and testability across services, restructuring poorly tested code to support proper unit testing
- Mentored and onboarded new hires through pair programming, code reviews, architecture walkthroughs, documentation

Security Hub *The Hut Group - Software Engineer (intern to mid)*

Jul 2018 - May 2020

- Joined as an intern on the Application Security team and grew into a core team member with increasing ownership over two years and onboarding over half of a team that grew from 3 to 10
- Led frontend development and contributed heavily to the backend of a greenfield role-based security management platform serving the CTO, CISO, and internal red/blue teams
- Angular frontend, Java/Kotlin (Spring Boot) backend, vulnerability scan orchestration via ActiveMQ, persistence in PostgreSQL, email reporter in Go
- Integrated nmap-based network scanning and OpenVAS for CVE scanning, with scheduled runs and manual re-run triggers per resource
- Role-based views driven by Windows Active Directory team assignments via Keycloak, scoping each stakeholder's visibility to their assigned VMs, networks, and domains
- Deployed and maintained on self-managed GKE: spun up clusters, maintained Helm charts, managed rolling deployments
- Platform has been in continuous production use since 2018, outlasting my tenure at the company
- Co-built (50/50) a high-throughput Content Security Policy violation reporting pipeline across THG's e-commerce estate, in Go, deployed on self-managed Kubernetes with Grafana for statistics and alerting - MVP in 2 weeks; production-ready in 6

Dotable *The Hut Group - Software Engineer*

Jun 2020 - Dec 2020

- Built a Website Builder SaaS as part of the Cloud Hosting Team
- Contributed backend integrations for a drag-and-drop website builder SaaS: domain registrar APIs (one via raw TCP with a custom command protocol, one XML-based) and Stripe subscription billing
- Microservices in Go and Kotlin; frontend in Gatsby and React

Core Security *Improbable - Software Engineer*

Jul 2021 - Dec 2021

- Built AWS Lambda-based Python integrations connecting Whitesource (Mend.io) and Cortex XDR to Jira, automatically generating security tickets based on vulnerability criteria - adopted in production
- Improved visibility of invalid/expired TLS certificates by surfacing upcoming anomalies as Jira tickets prior to automated renewal being in place

EDUCATION

University of Manchester - *September 2016 - June 2020*

- BSc in Computer Science (with Industrial Experience)

First Class Degree

OTHER

Hobbies & Interests

Electronics, 3D Printing, 3D Design, Puzzles, History